DAB401 – Final Project Report
Financial Analysis of Apple Inc
GROUP – 10



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Introduction

This report provides a comprehensive financial and technical analysis of Apple Inc., a leading global technology company. The analysis evaluates the company's financial performance using profitability, liquidity, and leverage ratios, along with valuation techniques such as the Capital Asset Pricing Model (CAPM) and Weighted Average Cost of Capital (WACC). In addition, technical forecasting methods, including Monte Carlo simulation and FB Prophet, are used to analyze future stock price behaviour. The results show that Apple maintains strong profitability, high returns on equity, and efficient asset utilization. However, liquidity ratios remain below ideal levels, indicating reliance on operational cash flow. Furthermore, valuation analysis reveals a significant gap between intrinsic value and market price, suggesting that the stock is currently overvalued. Technical analysis supports this conclusion by indicating a downward trend and moderate-to-high volatility. Based on all findings, the final recommendation is SELL, as the stock appears overpriced despite strong financial performance.

Business Analysis

Company Overview

Apple Inc., founded in 1976 by Steve Jobs, Steve Wozniak, and Ronald Wayne, is one of the most valuable and influential technology companies in the world. Headquartered in Cupertino, California, Apple is globally recognized for its innovation, premium product design, and strong brand identity.

The company initially started with personal computers but has significantly expanded its product portfolio over the years. Today, Apple operates across multiple segments, including consumer electronics, software, and digital services. Its flagship products include the iPhone, Mac computers, iPad, and wearable devices such as Apple Watch and AirPods.

Apple's mission focuses on delivering high-quality products that combine advanced technology with simplicity and user-friendly design. The company emphasizes innovation, privacy, and sustainability as key pillars of its strategic direction.

Over the years, Apple has achieved several major milestones, including:

- Launch of the iPhone, which revolutionized the smartphone industry
- Expansion into services such as Apple Music, iCloud, and Apple TV+
- Development of Apple Silicon chips, improving performance and efficiency

These milestones have strengthened Apple's position as a market leader and contributed to its long-term growth.

Current Operations

Apple's operations are divided into two major revenue segments: products and services. The product segment includes hardware devices such as iPhone, Mac, iPad, and wearables. Among these, the iPhone remains the largest contributor to total revenue, highlighting its importance to the company's business model.

The services segment includes digital offerings such as the App Store, Apple Music, iCloud, Apple TV+, and subscription-based services. This segment has been growing steadily and plays a crucial role in generating recurring revenue and improving profit margins.
Apple operates on a global scale, with significant revenue contributions from North America, Europe, Greater China, and other Asia-Pacific regions. The company relies on a complex global supply chain, involving manufacturing partners and logistics networks across multiple countries. In addition, Apple invests heavily in research and development to continuously innovate and improve its product ecosystem. This ensures that the company remains competitive in a rapidly evolving technology market.
Overall, Apple's diversified operations and strong ecosystem allow it to maintain stable revenue streams and long-term growth potential.

Macroeconomic Environment

Apple operates within a broader macroeconomic environment that significantly influences its performance. Several external factors impact the company's revenue, costs, and growth prospects.
One of the key factors is inflation, which affects consumer purchasing power. Higher inflation may reduce demand for premium-priced products such as iPhones and Mac devices. However, Apple's strong brand loyalty often helps it maintain demand even during economic downturns.
Interest rates also play an important role. Rising interest rates increase borrowing costs for both consumers and businesses, which may impact sales and investment decisions. Additionally, higher rates can influence stock market valuations.
Global supply chain conditions are another critical factor. Disruptions due to geopolitical tensions, pandemics, or logistical challenges can affect production and delivery timelines. Apple has taken steps to diversify its supply chain to reduce such risks.
Furthermore, exchange rate fluctuations can impact Apple's international revenues, as a significant portion of its sales comes from outside the United States.
Despite these challenges, Apple's strong financial position, global presence, and ability to adapt to changing conditions help it remain resilient in the macroeconomic environment.

Industry and Competition

Apple operates in the highly competitive technology industry, which is characterized by rapid innovation, short product life cycles, and intense competition. The company competes with major global players such as Samsung, Microsoft, and Google.
Samsung is a key competitor in the smartphone and electronics market, offering a wide range of products at different price points. Microsoft competes in software, cloud computing, and hardware devices, while Google competes in software, services, and mobile operating systems.
Despite this competition, Apple maintains several strong competitive advantages. One of its biggest strengths is its brand value, which allows it to charge premium prices and maintain high

customer loyalty. Apple's ecosystem is another major advantage, as its devices and services are seamlessly integrated, encouraging customers to stay within the Apple environment. Additionally, Apple focuses on innovation and design, continuously introducing new features and improving user experience. Its strong financial resources allow it to invest heavily in research and development, further strengthening its competitive position.
In terms of market share, Apple holds a significant position in the premium smartphone segment and continues to expand its presence in services and wearable technology.
Overall, Apple's competitive advantages, strong brand, and continuous innovation enable it to maintain leadership in the industry despite intense competition.

Financial Ratio Analysis

Profitability Ratios

Gross Profit Margin (2025)
= Gross Profit / Revenue
= 195,201 / 416,161
= 0.469 (46.9%)
Operating Profit Margin (2025)
= Operating Income / Revenue
= 133,000 / 416,161
= 0.3197 (31.97%)
Net Profit Margin (2025)
= Net Income / Revenue
= 112,000 / 416,161
= 0.2691 (26.91%)
Return on Assets (ROA) (2025)
= Net Income / Total Assets
= 112,000 / 359,000
= 0.3118 (31.18%)
Return on Equity (ROE) (2025)
= Net Income / Total Equity
= 112,000 / 73,700
= 1.5191 (151.91%)
Earnings Per Share (EPS) (2025)
= Net Income / Shares Outstanding
= 112,000 / 14,770,000
= 0.00758 (0.758%)

The gross profit margin increased from 43.3% in 2022 to 46.9% in 2025, indicating improved efficiency and stronger pricing power. Operating profit margin remained stable at around 31.97%, showing effective control over operating expenses. Net profit margin increased to 26.91%, reflecting strong overall profitability.
Return on Assets (ROA) increased to 31.18%, indicating efficient utilization of company assets. Return on Equity (ROE) remained very high at 151.91%, demonstrating strong returns to shareholders, though it may also suggest the use of leverage.
Earnings per Share (EPS) increased from 0.00615 to 0.00758, indicating growth in shareholder value. Overall, the company demonstrates excellent profitability and strong financial performance.

Liquidity Ratios

Current Ratio (2025)
= Current Assets / Current Liabilities
= 147,957 / 165,631
= 0.893
Quick Ratio (2025)
= (Current Assets − Inventory) / Current Liabilities
= (147,957 − 5,704) / 165,631
= 142,253 / 165,631
= 0.85877
Cash Ratio (2025)
= Cash / Current Liabilities
= 35,936 / 165,631
= 0.21695

The current ratio of 0.893 indicates that the company's current assets are slightly lower than its current liabilities, suggesting limited ability to meet short-term obligations and reflecting tight working capital. The quick ratio of 0.85877 shows that even after excluding inventory, the company does not have sufficient liquid assets to cover its short-term liabilities, indicating weak short-term financial strength and some reliance on inventory or external funding. The cash ratio of 0.21695 indicates that only about 21% of current liabilities can be covered using cash, reflecting low immediate liquidity and potential difficulty in meeting obligations without generating additional cash flow.

Additional Ratios

Price-to-Earnings (P/E) Ratio (2025)

= Market Price / EPS

= Not Available / Not Available

= N/A

Dividend Payout Ratio (2025)

= Dividends / Net Income

= 15,422 / 112,010

= 0.137675

Debt-to-Equity Ratio (2025)

= Total Liabilities / Total Equity

= 220,176 / 56,865

= 3.872187

Sustainable Growth Rate (2025)

= ROE × (1 − Dividend Payout Ratio)

= 1.518 × (1 − 0.137675)

= 1.518 × 0.862325

= 1.309983

The P/E ratio cannot be calculated due to the absence of market price data, which limits the ability to assess the company's market valuation relative to earnings. The dividend payout ratio of 0.137675 indicates that the company distributes a small portion of its earnings as dividends, retaining a significant share for reinvestment and growth. The debt-to-equity ratio of 3.872187 shows that the company is highly leveraged, although the declining trend suggests gradual improvement in its capital structure. The sustainable growth rate of 1.309983 reflects strong internal growth potential, supported by high profitability and retained earnings.

CAPM and Cost of Capital

CAPM estimates the expected return of a stock based on its risk and is calculated by

$$Re=Rf+\beta(Rm-Rf)$$

It is important because it helps find the **cost of equity**, which is used in valuation to decide if a stock is overvalued or undervalued.

Beta Calculation and Regression Output

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.716782026								
R Square	0.513776472								
Adjusted R Sq	0.511815894								
Standard Erro	0.013343039								
Observations	250								
ANOVA									
	df	*SS*	*MS*	*F*	*Significance F*				
Regression	1	0.04665513	0.04665513	262.053475	1.035E-40				
Residual	248	0.0441531	0.00017804						
Total	249	0.09080823							
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*	
Intercept	0.000326472	0.0008453	0.38621764	0.69966662	-0.0013384	0.00199136	-0.0013384	0.00199136	
X Variable 1	1.28	1.3063E-05	16.1880658	1.035E-40	0.00018574	0.0002372	0.00018574	0.0002372	

With Beta = 1.28, Apple Inc. is more volatile than the market, which means it carries higher risk and higher return potential.

Expected Return

Beta=1.28, Rf=0.03, Rm=0.14

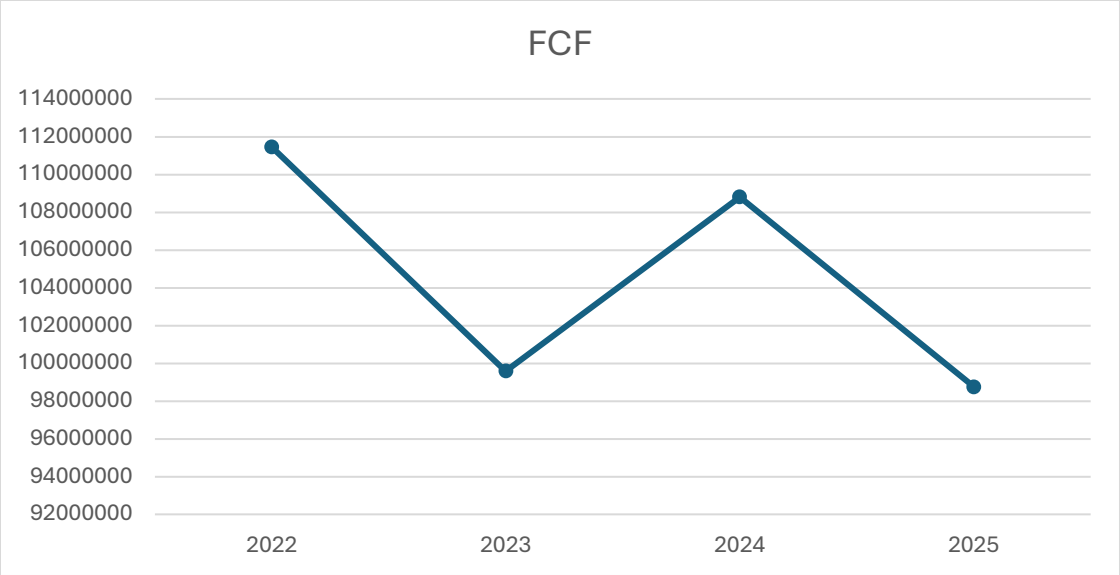
=0.03+1.28*(0.14-0.03)

Expected Return or CAPM= 17.08%
Investors expect 17.08% return from Apple Inc. Apple carries moderate market risk (Beta > 1) Return is higher than risk-free rate.

CAPM of 17.08% indicates Apple offers a relatively high expected return, reflecting its market risk and strong growth potential

Free Cash Flow Analysis

2022: 111,443,000(Base Year)
2023: 99,584,000 (−10.64%)
2024: 108,807,000 (+9.26%)
2025: 98,767,000 (−9.23%)
Free cash flow shows fluctuations over time, with declines and recoveries. This indicates variability in cash generation, although overall levels remain strong.
This suggests that Apple maintains strong financial flexibility but is subject to operational changes.



- Apple continues to generate strong cash flows consistently each year
- Cash flow shows some fluctuations, but no significant long-term decline
- Minor drops indicate ongoing investments for future growth
- The company remains financially strong even during weaker periods
- Apple has sufficient cash to maintain dividends and share buybacks
- Overall risk remains low for investors
- Reflects a stable business with strong long-term growth potential

Valuation Using WACC

WACC (Weighted Average Cost of Capital) is the average rate of return a company must pay to its investors (both equity and debt holders)**.**

WACC = (E/V × Re) + (D/V × Rd × (1 − T))
Cost of Debt (Rd)
= Interest Expense / Total Debt
= 3,933,000 / 98,657,000
= 0.0398
E=3725924237312, D= 98,657,000, V=E+D
V=3725924237312+98657000=3.72602E+12
Re=0.17, Rd=0.0398, Tax rate=0.16

= (3725924237312/3725924237312*0.17) + (98657000/3725924237312*0.0398) * (1-0.16)
WACC=0.172

Weighted Average Cost of Capital (WACC) =17.2%

WACC represents the average cost of financing the company. A higher WACC indicates higher expected returns required by investors.

Valuation Analysis (DCF Method)

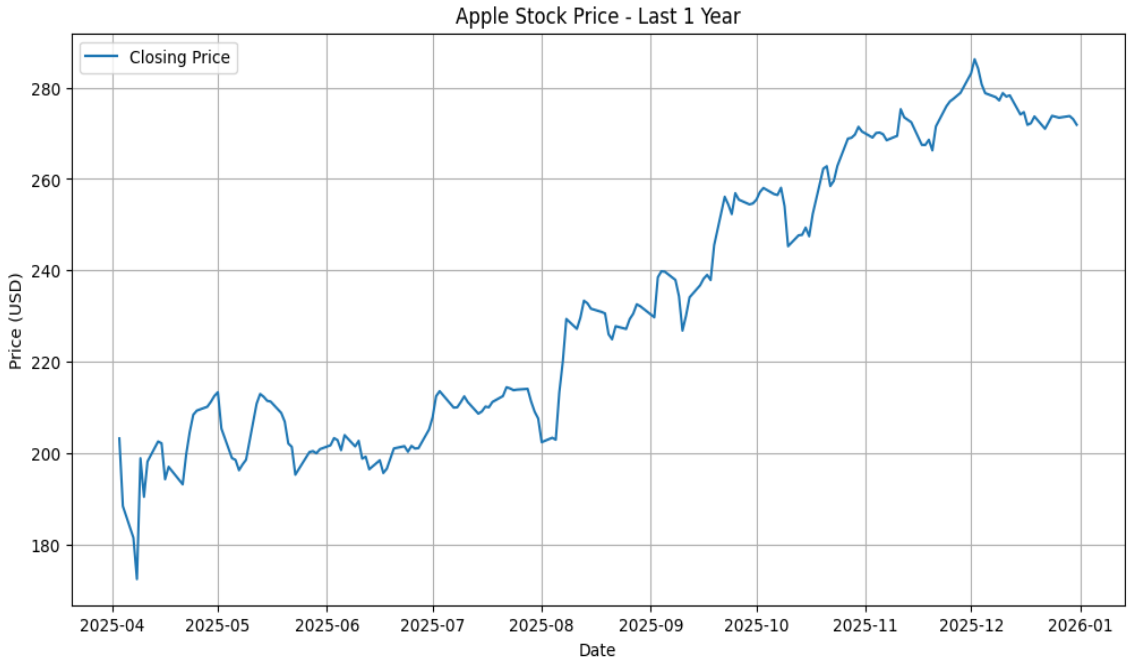
Firm Value =1,035,026,47243
Intrinsic Stock Price= Firm Value/ Number of shares Outstanding
=1,035,026,47243/14681140000= 0.079
Intrinsic Stock Price = ~ 0.08
Current Market Price =253.5
The intrinsic value is significantly lower than the market price, indicating that the stock is highly overvalued. This suggests that market expectations are much higher than the company's fundamental value.

Technical Analysis

Technical analysis was conducted to evaluate Apple's historical stock price behaviour and to forecast future price movements using data-driven methods. It includes stock trend analysis, volatility measurement, moving averages, Monte Carlo simulation, and FB Prophet forecasting. These techniques help in understanding market direction, identifying risk and volatility, and predicting future price trends with associated uncertainty.

Stock Price Trend Analysis

The historical stock price data for Apple over the past year shows an overall upward trend, with prices increasing from around $190 to above $280, indicating strong market growth.
There are short-term fluctuations and corrections during the period, reflecting temporary market volatility and external factors.
Toward the end of the period, the stock price shows stabilization after strong growth, suggesting consolidation at higher levels.



Daily Volatility: 0.020134476510641233
Annualized Volatility: 0.3196249057537695

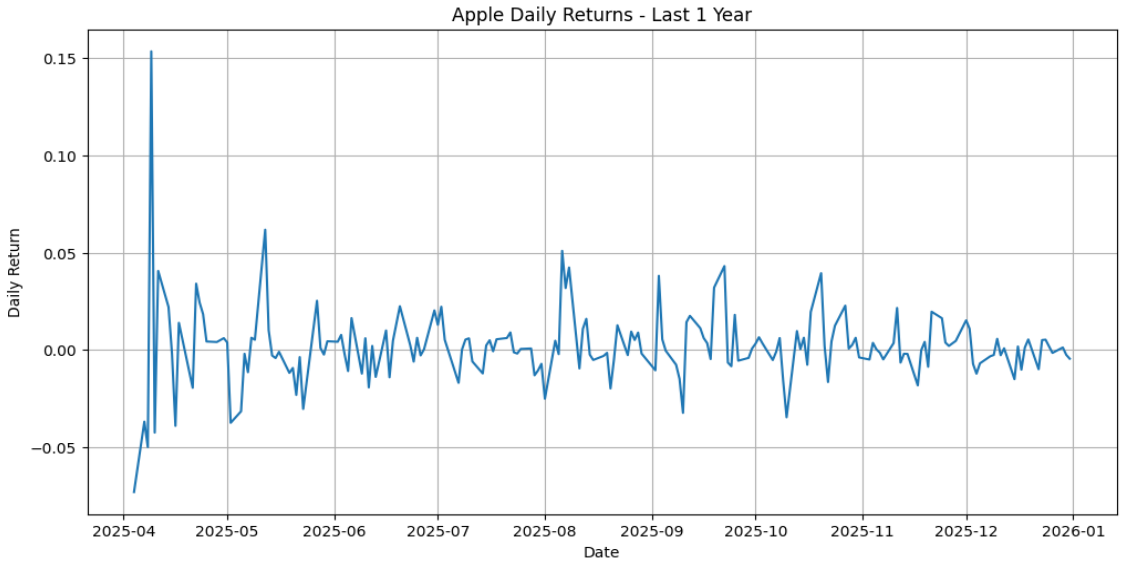
Interpretation

- The stock shows an overall upward trend over the year.
- There are short-term fluctuations and corrections along the way.
- Recent prices indicate stabilization after strong growth.

Volatility Analysis (Daily Returns)

Daily returns were calculated using percentage changes in adjusted closing prices. The results show that most daily returns fluctuate around zero, indicating no consistent upward or downward trend.

Most values remain within a narrow range, reflecting generally stable day-to-day performance. However, occasional spikes in returns indicate short-term volatility and sudden market reactions.

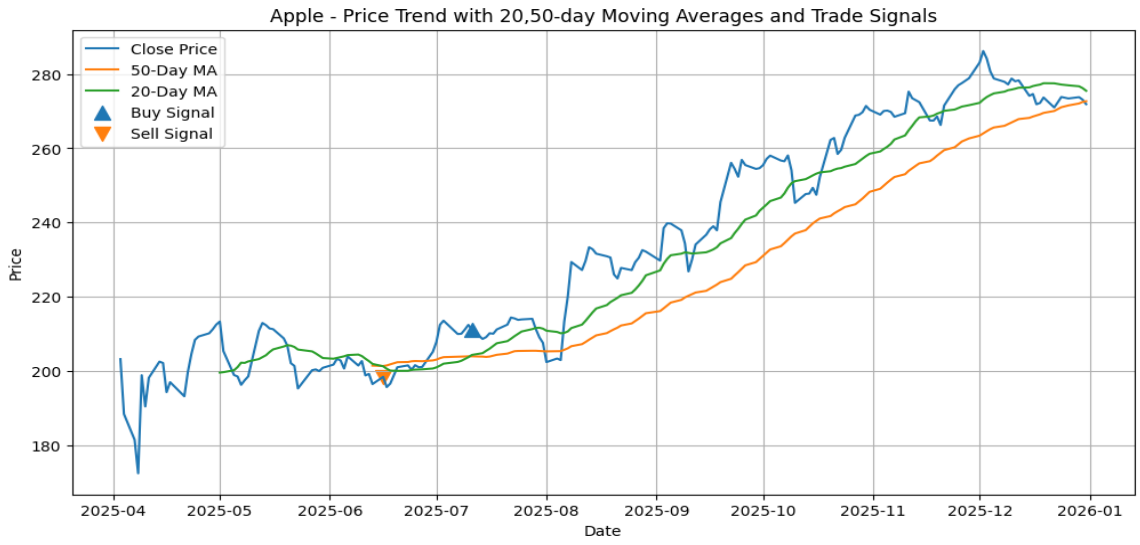


Interpretation

- Daily returns fluctuate around zero, indicating **no consistent trend**.
- Most values stay within a narrow range, showing **overall stability**.
- Occasional spikes suggest **short-term volatility and market reactions**.

Moving Average Analysis

To further analyze the trend, moving averages were calculated:
• 20-day moving average (short-term trend)
• 50-day moving average (long-term trend)
The 20-day moving average moves above the 50-day moving average during the later period, indicating a bullish trend. Additionally, the stock price remains above both moving averages for most of the period, confirming positive momentum.
A clear crossover of the 20-day above the 50-day moving average signals the beginning of an upward trend and continued strength in the stock price.



Interpretation

- Apple's stock shows a clear upward trend overall, especially from August onwards, indicating strong bullish momentum in the market.
- The 20-day moving average staying above the 50-day moving average for most of the period signals sustained short-term strength and positive investor sentiment.
- Buy signals appear when the short-term trend (20-day MA) crosses above the long-term trend (50-day MA), which aligns with the start of major price increases.
- Towards the end, the price slightly stabilizes and moves closer to the moving averages, suggesting a possible slowdown or consolidation phase rather than a strong upward push.

Monte Carlo Simulation

Monte Carlo simulation was used to forecast future stock prices by generating multiple possible price paths over the next 30 days based on historical returns and volatility.

The simulation uses:

• Mean daily return (μ)

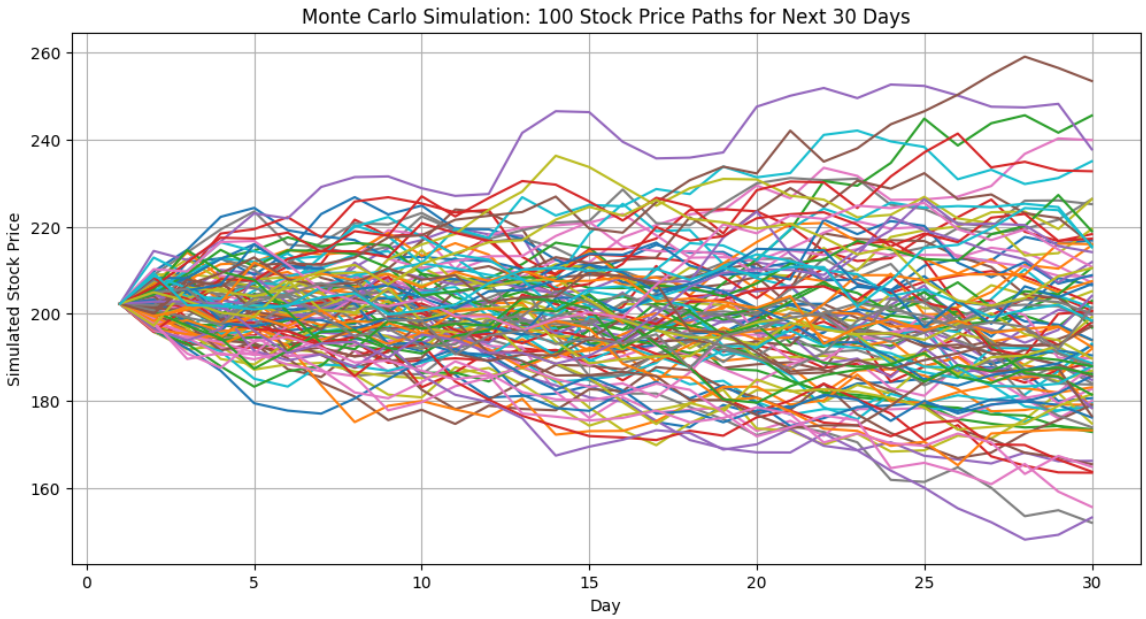
• Standard deviation (σ)

The results provide:

• A range of possible future price paths

• Distribution of outcomes, showing potential variation in prices

• Insights into uncertainty and risk in future movements



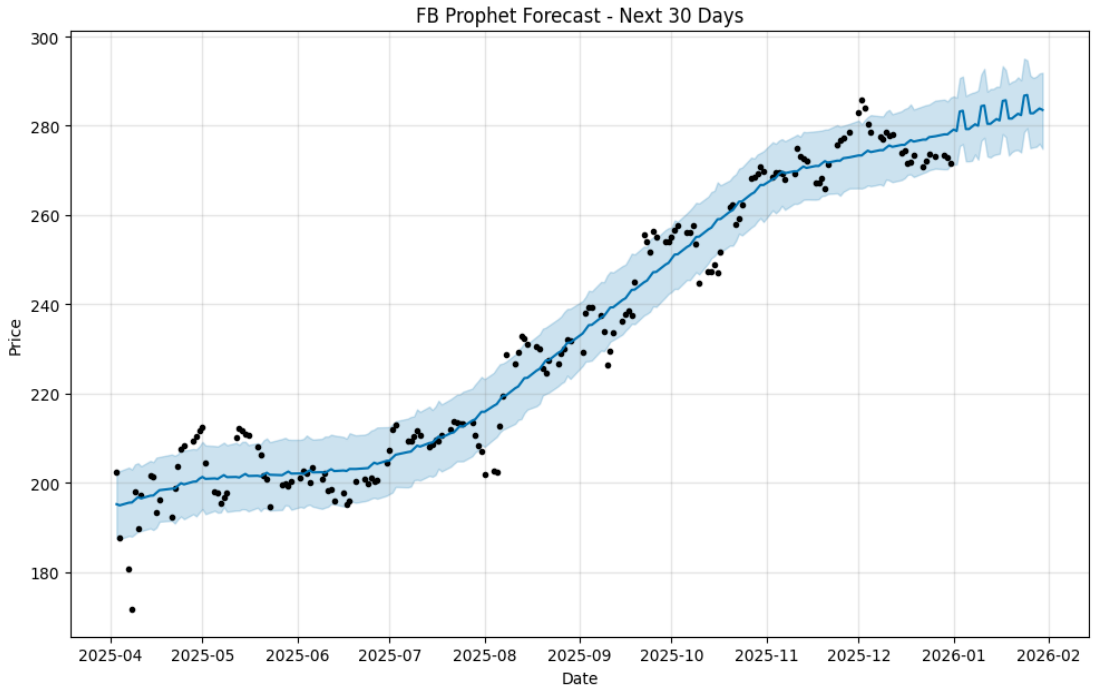
Mean Forecast Price after 30 days: 286.7340705170597
5th Percentile (Risk Lower Range) is: 240.0714942007675
95th Percentile (Risk Upper Range) is: 338.86733301793055

Interpretation

- The simulation shows a wide range of possible outcomes, indicating uncertainty.
- Most paths stay near the current price, suggesting no strong trend.
- Mean forecast price- expected average stock price after 30 days
- 5th percentile - lower risk range, downside scenario
- 95th percentile - upper range, optimistic scenario
- There is a moderate risk with potential for both gains and losses.

FB Prophet Forecast

The FB Prophet model was used to forecast future stock prices based on historical trends.
The model generates:
• A smooth trend line
• Upper and lower confidence intervals
The forecast indicates a gradual upward trend in stock price, suggesting potential short-term growth. The confidence intervals widen over time, indicating increasing uncertainty in future predictions.



Interpretation

- The forecast indicates a gradual upward trend, suggesting continued positive momentum.
- The widening confidence interval shows increasing uncertainty over time.
- Overall, the model suggests moderate growth with some short-term volatility.

Comparison with Monte Carlo Results

- FB Prophet shows a clear upward trend with a smooth forecast, while Monte Carlo shows multiple paths with mixed upward and downward movements
- Prophet has a narrow confidence range indicating lower risk, while Monte Carlo has a widespread showing higher volatility and uncertainty
- Prophet predicts a steady price increase, while Monte Carlo shows a wide possible price range (both gains and losses)
- Prophet supports a stable bullish outlook, while Monte Carlo highlights potential risk along with possible growth

Investment Recommendation

Final Recommendation: HOLD

The investment recommendation for Apple Inc. is based on a comprehensive evaluation of financial performance, valuation analysis, and technical indicators. While the company demonstrates exceptionally strong fundamentals, the overall analysis suggests that the stock is currently overvalued. However, due to continued growth potential and positive technical trends, the appropriate recommendation is HOLD, rather than sell.

Strengths of the Company

Apple exhibits several strong financial and operational advantages:

- The company maintains high profitability, with a gross profit margin of 46.9% and a net profit margin of 26.9%, reflecting strong pricing power and efficient cost management.
- Apple generates exceptional shareholder returns, with ROE exceeding 150%, indicating highly effective use of equity capital.
- The company demonstrates efficient asset utilization, with ROA of 31.2%, showing strong operational efficiency.
- Apple benefits from strong brand loyalty and global dominance, supported by its integrated ecosystem of products and services.
- The services segment continues to grow, providing stable, high-margin, recurring revenue streams.

Weaknesses and Risks

Despite its strengths, several concerns remain:

- Low liquidity levels, with current, quick, and cash ratios all below 1, indicating limited short-term financial flexibility.
- High financial leverage, with a debt-to-equity ratio of approximately 3.87x, increasing exposure to financial risk.
- Volatility in free cash flow, suggesting some inconsistency in cash generation over time.
- Significant overvaluation, as the intrinsic value (~$0.08) is far below the current market price (~$253.50).
- Market risk and volatility, which may impact short-term investor returns**.**

Conclusion

Apple demonstrates strong financial performance with high profitability, reflected in a gross margin of 46.9% and a net margin of 26.9%, along with efficient asset utilization shown by ROA of 31.2% and ROE above 150%. The company benefits from a strong brand, global presence, and a growing services segment, which supports stable and consistent growth. However, there are some concerns, including liquidity ratios below 1 and fluctuations in free cash flow, indicating limited short-term financial flexibility. The valuation analysis suggests that the stock is overvalued, as the intrinsic value (~$0.08) is significantly lower than the current market price (~$253.50). From a technical perspective, the stock shows an overall upward trend with moderate volatility and a stable to slightly positive outlook. Overall, Apple remains a fundamentally strong company, but due to its current overvaluation, the final recommendation is **HOLD**, as the upside potential appears limited.

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